

Real Analysis I — Homework 5

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Unless otherwise specified, all functions are defined on $E \subset \mathbb{R}^n$ where $E \in \mathcal{M}$.

For notation simplicity, a set such as $\{x \in E : f(x) > t\}$ is written as $\{f > t\}$ for short.

1. Suppose $f \in L(E)$. Let E_k be non-decreasing and $\lim_k E_k = E$. Show that

$$\lim_{k \rightarrow \infty} \int_{E_k} f = \int_E f.$$

2. Suppose $f \in L(E)$ where $\mu(E) < \infty$. Let $E_{k,j} = \{\frac{j}{k} \leq f < \frac{j+1}{k}\}$. Show that

$$\lim_{k \rightarrow \infty} \sum_{j=-\infty}^{\infty} \frac{j}{k} \cdot \mu(E_{k,j}) = \int_E f.$$

3. Suppose $f \in L(\mathbb{R})$ and $f(0) = 0$, and f is differentiable at $x = 0$. Show that $\frac{f(x)}{x} \in L(\mathbb{R})$.

4. Let $f \in C([a, b])$ (i.e., $f : [a, b] \rightarrow \mathbb{R}$ is continuous). Show that

$$\lim_{k \rightarrow \infty} \int_a^b f(x) \sin(kx) dx = 0.$$

Hint: For any $\epsilon > 0$, consider a proper partition $\Delta : a = x_0 < \dots < x_n = b$ and $K \in \mathbb{N}$ such that the absolute value of the right hand side of

$$\int_a^b f(x) \sin(kx) dx = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} (f(x) - f(x_i)) \sin(kx) dx + \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x_i) \sin(kx) dx$$

can be bounded by ϵ for all $k \geq K$.

5. Use the fact in the previous problem to show that the claim is true if $f \in L([a, b])$.
6. Suppose $\mu(E) < \infty$ and $f_k : E \rightarrow \mathbb{R}$ are measurable. Show that $f_k \xrightarrow{\mu} 0$ on E iff

$$\lim_{k \rightarrow \infty} \int_E \frac{|f_k|}{1 + |f_k|} = 0.$$

7. Suppose $f : E \rightarrow \mathbb{R}$, and for any $\epsilon > 0$, there exist $g, h \in L(E)$ such that $g \leq f \leq h$ and

$$\int_E |h - g| \leq \epsilon.$$

Show that f is measurable on E and $f \in L(E)$.

8. Suppose $f_k, f : E \rightarrow \mathbb{R}$ are measurable, $f_k \rightarrow f$ a.e., and there exists $M > 0$ such that

$$\int_E |f_k| \leq M$$

for all k . Show that $f \in L(E)$.

9. Let $F \subset [0, 1]$ be closed and $\mu(F) = 0$. Show that χ_F is Riemann integrable on $[0, 1]$.
10. Suppose $f : [0, 1] \rightarrow [a, b]$ is Riemann integrable, $g \in C([a, b])$. Show that $g \circ f$ is Riemann integrable on $[0, 1]$.